

Lab 6 Deliverable

```
## [1] "Author: Sebastian Hyland, 39878921"
```

First, let us load in the data and perform a logistic regression with Lag1 + Lag2 + Lag3. Binomial family used since Direction is a binary outcome.

```
library(ISLR)

glm.fit <- glm(Direction ~ Lag1 + Lag2 + Lag3,
  data = Smarket,
  family = binomial
)

summary(glm.fit)

##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3, family = binomial,
##      data = Smarket)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.074230   0.056672   1.310   0.190
## Lag1        -0.071404   0.050102  -1.425   0.154
## Lag2        -0.044260   0.050019  -0.885   0.376
## Lag3         0.008873   0.049855   0.178   0.859
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1731.2  on 1249  degrees of freedom
## Residual deviance: 1728.4  on 1246  degrees of freedom
## AIC: 1736.4
##
## Number of Fisher Scoring iterations: 3
```

Now, we can make predictions and determine the confusion matrix

```
glm.probs <- predict(glm.fit, type = "response")
glm.pred <- ifelse(glm.probs > 0.5, "Up", "Down")

confusion_matrix <- table(glm.pred, Smarket$Direction)
print(confusion_matrix)

##
## glm.pred Down  Up
##      Down  114  97
##      Up   488 551
```

And finally calculate classification rate (accuracy)

```
classification_rate <- mean(glm.pred == Smarket$Direction)
print(classification_rate)
```

```
## [1] 0.532
```